

Tejasva Singh Chouhan

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EDUCATION

Shri G. S. Institute of Technology and Science, Indore

2024 – 2028

B.Tech in Computer Engineering (CGPA: 8.22)

Indore, India

- Relevant Coursework: Structured Programming, Data Structures, Design and Analysis of Algorithms, Database Management Systems, Microprocessors

EXPERIENCE

NIT Bhopal

Present

Research Intern – Exploration of VLM for Handwritten Text Recognition Under the guidance of a PhD Scholar, IIT Bombay

- Conducting research on Vision Language Models (VLMs) for handwritten text recognition with a focus on multilingual and Indian-language documents.
- Evaluating state-of-the-art OCR and VLM systems including PaddleOCR, EasyOCR, TrOCR, and multimodal transformer-based architectures.
- Performing comparative analysis using CER and WER metrics on handwritten text datasets.

IndhanPay Pvt. Ltd.

2025

Backend & AI Automations Intern

Remote

- Completed an 85-day internship focused on backend development and AI-driven automation workflows.
- Developed backend APIs and automation scripts to streamline internal processes.
- Worked with Node.js systems integrating authentication, databases, and external APIs.
- Received appreciation for reliability and timely delivery of development tasks.

E-NOTEBOOK

2025

Backend & Database Development Intern

Remote

- Led backend and database development for the E-NOTEBOOK-8 platform, delivering scalable APIs, optimized database schemas, and production-ready system improvements.

PROJECTS

Cognitive-State-Aware Adaptive AI Agent System | *Next.js, FastAPI, LangGraph, Ollama*

2026

- Developed a cognitive-state-aware multi-agent system that dynamically adapts agent workflows based on user workload levels.
- Built a FastAPI backend using LangGraph with Planner, Research, Verifier, and Summarizer agents.
- Implemented adaptive routing policies for NORMAL, MODERATE, HIGH, and OVERLOADED workload states.
- Designed a Next.js dashboard visualizing agent execution paths, workload states, and system metrics in real time.

Custom Database Engine with B+ Tree Indexing | *Java, Spring Boot*

2025

- Built a custom database engine with a fully functional B+ Tree supporting fast lookups, inserts, deletes, and range queries.
- Implemented DBMS internals including node split/merge, file-based persistence, caching, and block storage.
- Designed a REST API layer where CRUD operations were executed directly through the custom B+ Tree.
- [GitHub Repository](#)

AI Code Generation Workflow CLI | *TypeScript, Node.js*

2025

- Built a CLI tool orchestrating AI-assisted code generation with structured planning, review, approval, and verification workflows.
- Implemented modular commands using Commander.js with interactive and batch execution modes.
- Designed a staging system with diff-based review and selective application of generated code changes.
- [GitHub Repository](#)

TECHNICAL SKILLS

Languages: C, C++, JavaScript, TypeScript, SQL, HTML, CSS

Frontend: React.js, Next.js, Tailwind CSS, Responsive Web Design, UI/UX Principles

Backend: Node.js, Express.js, FastAPI, REST APIs

Databases: PostgreSQL, Appwrite, Knex.js, Database Schema Design

AI/ML: LangGraph, LangChain, Ollama, Vision Language Models (VLMs), OCR Evaluation

Tools & Technologies: Git, GitHub, JWT, Bcrypt, Nodemailer, Zoho API, Docker, Netlify